# WORKSHEET — UNIT II

## IoT Communication and Networking

**Course:** IoT with Applications

**Unit:** II (3 Hrs / 180 min)

**Topics:** Short-Range Communication; Long-Range Communication; IoT Communication Protocols; IoT Networking Concepts; Comparison of Technologies

## 🎯 Section A — Learning Objectives

**1.** Explain short-range communication technologies — Bluetooth/BLE, Zigbee, RFID, NFC and Wi-Fi — and their typical uses.

**2.** Describe long-range technologies — LoRaWAN, NB-IoT, LTE-M and 4G/5G — and choose the right one for a given application.

**3.** Explain the MQTT publish-subscribe model (broker, topics) and compare it with CoAP, HTTP/HTTPS, AMQP and WebSockets.

**4.** Describe IoT networking concepts: IPv4 vs IPv6, 6LoWPAN, Wireless Sensor Networks, gateways, addressing and routing.

**5.** Compare communication technologies on range, power consumption, bandwidth and use-case.

**6.** Select a suitable communication technology and protocol for a real-world IoT scenario.

## 🔑 Section B — Key Concepts

| No. | Concept | Short Explanation |
|---|---|---|
| 1 | **Short-Range vs Long-Range** | Short-range links (centimetres to ~100 m) suit local device networks; long-range links (kilometres) suit wide-area, low-power deployments. |
| 2 | **Bluetooth Low Energy (BLE)** | A low-power version of Bluetooth for short-range links between devices such as wearables, beacons and phones. |
| 3 | **Zigbee** | A low-power, low-data-rate mesh networking standard based on IEEE 802.15.4, widely used in home and industrial automation. |
| 4 | **LPWAN (LoRaWAN, NB-IoT, LTE-M)** | Low-Power Wide-Area Networks send small amounts of data over long distances while letting battery-powered devices last for years. |
| 5 | **Publish-Subscribe (MQTT)** | Clients publish messages to topics on a broker; the broker forwards them to every client subscribed to that topic. |
| 6 | **Request-Response (HTTP, CoAP)** | A client sends a request to a server and waits for a reply; CoAP is a lightweight, UDP-based version for constrained devices. |
| 7 | **IPv6 in IoT** | A 128-bit addressing scheme that gives every device a unique address, solving the IPv4 address shortage. |
| 8 | **6LoWPAN** | Lets IPv6 packets travel over low-power IEEE 802.15.4 networks using header compression and fragmentation. |
| 9 | **Gateway** | A device that connects local IoT networks to the internet and translates between different protocols and networks. |

## 📘 Section C — Important Terminology

| Keyword | Meaning |
|---|---|
| **Bluetooth / BLE** | Short-range wireless standard in the 2.4 GHz band; BLE is the low-energy variant for battery-powered devices. |

| | |
|---|---|
| **Zigbee** | Low-power mesh networking standard built on IEEE 802.15.4 for monitoring and control applications. |
| **RFID** | Radio-Frequency Identification: tags respond to a reader's radio signal to identify and track objects. |
| **NFC** | Near-Field Communication: very short-range (a few centimetres) wireless link used for contactless payment and pairing. |
| **Wi-Fi** | IEEE 802.11 wireless networking offering high bandwidth over local areas, at higher power consumption. |
| **LoRaWAN** | Long-range, low-power network protocol built on LoRa radio that works in unlicensed sub-GHz bands. |
| **NB-IoT** | Narrowband IoT: a cellular low-power wide-area technology in licensed spectrum with deep indoor coverage. |
| **LTE-M** | Cellular LPWA technology with higher data rates and mobility support than NB-IoT. |
| **MQTT** | Lightweight publish-subscribe messaging protocol that uses a broker and topics, typically over TCP. |
| **CoAP** | Constrained Application Protocol: a lightweight RESTful protocol for constrained devices, typically over UDP. |
| **AMQP** | Advanced Message Queuing Protocol: a reliable, queue-based messaging protocol used in enterprise systems. |
| **WebSocket** | A protocol giving full-duplex, persistent two-way communication over a single TCP connection. |
| **6LoWPAN** | IPv6 over Low-Power Wireless Personal Area Networks: adapts IPv6 for small, low-power radio links. |
| **WSN** | Wireless Sensor Network: many spatially distributed sensor nodes that cooperate to monitor an area. |
| **Gateway** | A bridge between local IoT networks and the internet or cloud that can translate protocols. |

## ✅ Section D — Multiple-Choice Questions [Easy–Medium]

**1. Which short-range technology is designed for very low power consumption and is widely used in wearables and beacons?**

**A.** Wi-Fi
**B.** Bluetooth Low Energy (BLE)
**C.** LoRaWAN
**D.** NB-IoT

**2. Zigbee is based on which IEEE standard?**

**A.** IEEE 802.3
**B.** IEEE 802.11
**C.** IEEE 802.15.4
**D.** IEEE 802.16

**3. What is the typical working range of NFC?**

**A.** About 100 m
**B.** About 10 km
**C.** About 1 km
**D.** A few centimetres

**4. Which long-range technology typically operates in unlicensed sub-GHz spectrum?**

**A.** LoRaWAN
**B.** NB-IoT
**C.** LTE-M
**D.** 5G

**5. Which cellular LPWA technology offers higher data rates and supports mobility, making it suitable for asset tracking?**

**A.** NB-IoT
**B.** LTE-M
**C.** Zigbee
**D.** NFC

**6. In MQTT, which component receives published messages and forwards them to subscribers?**

**A.** Topic
**B.** Sensor
**C.** Broker
**D.** Gateway

**7. Which protocol is designed for constrained devices, uses UDP, and follows a RESTful request-response model?**

**A.** MQTT
**B.** AMQP
**C.** HTTP
**D.** CoAP

**8. Which protocol provides full-duplex, persistent two-way communication over a single TCP connection?**

**A.** WebSockets
**B.** CoAP
**C.** NFC
**D.** Zigbee

**9. How many bits are there in an IPv6 address?**

**A.** 32
**B.** 128
**C.** 64
**D.** 256

**10. What is the purpose of 6LoWPAN?**

**A.** To carry audio over Bluetooth
**B.** To replace Ethernet cables
**C.** To carry IPv6 packets over low-power IEEE 802.15.4 networks
**D.** To move LoRa into licensed spectrum

## ✏️ Section E — Fill in the Blanks [Easy]

**1.** MQTT follows a _____ model in which clients publish messages to topics and other clients subscribe to them.

**2.** A _____ connects local IoT devices to the internet and can translate between different protocols.

**3.** _____ is a cellular low-power wide-area technology that offers deep indoor coverage in licensed spectrum.

**4.** CoAP normally runs over _____ rather than TCP.

**5.** A _____ is a network of spatially distributed sensor nodes that cooperate to monitor physical conditions.

## ☑️ Section F — True or False [Easy]

**1.** Bluetooth Low Energy is designed to consume very little power. (True / False)

**2.** LoRaWAN is a short-range technology meant for connecting devices inside a single room. (True / False)

**3.** In MQTT, publishers send messages directly to subscribers without a broker. (True / False)

**4.** IPv6 provides a far larger address space than IPv4. (True / False)

**5.** WebSockets require a brand-new connection for every message that is sent. (True / False)

## 🔗 Section G — Match the Following [Medium]

| Column A | Column B (shuffled) |
|---|---|
| 1. MQTT | a. Cellular LPWA technology in licensed spectrum with deep indoor coverage |
| 2. CoAP | b. Lightweight publish-subscribe protocol that uses a broker |
| 3. LoRaWAN | c. Bridge between local IoT networks and the internet |
| 4. NB-IoT | d. IPv6 over low-power IEEE 802.15.4 networks |
| 5. Zigbee | e. Low-power mesh network standard based on IEEE 802.15.4 |
| 6. 6LoWPAN | f. Lightweight RESTful protocol that mainly uses UDP |
| 7. Gateway | g. Long-range, low-power protocol in unlicensed sub-GHz bands |

## 📝 Section H — Short-Answer Questions [Medium]

**1. Differentiate between short-range and long-range IoT communication technologies, with one example of each.**

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**2. What is the role of the broker in MQTT?**

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**3. How does CoAP differ from HTTP?**

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**4. Why is IPv6 important for IoT?**

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**5. What is the function of an IoT gateway?**

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## 📄 Section I — Descriptive Questions [Medium–Advanced]

**1. Compare Bluetooth/BLE, Zigbee, Wi-Fi, RFID and NFC in terms of range, power consumption, data rate and typical applications.**

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**2. Explain LoRaWAN, NB-IoT, LTE-M and 4G/5G for IoT, and state when you would choose each one.**

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**3. Explain the MQTT publish-subscribe model (broker, topics, publishers, subscribers) with a diagram, and compare MQTT with CoAP, HTTP/HTTPS, AMQP and WebSockets.**

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**4. Describe IPv4 and IPv6 in IoT, 6LoWPAN, Wireless Sensor Networks, gateways, and addressing and routing in IoT networks.**

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## 💻 Section J — Practical / Application-Based Questions [Advanced]

### Practical 1

Problem: Simulate the MQTT publish-subscribe idea in Python using a dictionary as a simple broker.

**Task:**

**1.** Create a dictionary that maps each topic name to a list of subscriber names.

**2.** Write subscribe(topic, name) and publish(topic, message) functions.

**3.** Subscribe two clients to the topic “home/temperature”, publish one message, and print which clients received it.

**Expected Output:** *One line per subscriber showing the message received on the topic; a topic with no subscribers prints nothing.*

### Practical 2

Problem: Choose a suitable communication technology for each scenario and justify your choice.

**Task:**

**1.** A fitness band sending heart-rate data to a phone, running for weeks on a small battery.

**2.** Hundreds of battery-powered soil sensors spread across a farm several kilometres wide, sending a few bytes every hour.

**3.** A contactless access card tapped on a door reader to open it securely.

**Expected Output:** *A table giving the technology, range, power need and a one-line reason for each scenario.*

### Practical 3

Problem: Design an MQTT topic structure for a two-floor smart building with temperature and light sensors in each room.

**Task:**

**1.** Write a clear topic hierarchy for the sensors (building → floor → room → sensor type).

**2.** Write a subscription that receives temperature from every room using a wildcard.

**3.** Choose a QoS level for routine readings and for a high-temperature alert, with a reason.

**Expected Output:** *A topic tree such as building/floor1/room101/temperature, a wildcard subscription, and a justified QoS choice for each message type.*

## 🗺️ Section K — Concept-Mapping Activity

**IoT COMMUNICATION AND NETWORKING**

*Complete the map below: fill in the missing sub-topic name or example for each branch.*

| Branch 1 | Branch 2 | Branch 3 | Branch 4 |
|---|---|---|---|
| **Short-Range** | **Long-Range** | **Protocols** | **Networking Concepts** |
| Example: _____ | Example: _____ | Example: _____ | Example: _____ |

*(Answer key — example values: BLE, Zigbee, Wi-Fi, NFC; LoRaWAN, NB-IoT, LTE-M; MQTT, CoAP, AMQP; IPv6, 6LoWPAN, gateway)*

## 🏢 Section L — Case Study / Real-World Activity

### Scenario

A farm covering several square kilometres wants to monitor soil moisture using about 200 battery-powered sensors. Each sensor sends a few bytes every 30 minutes, and the batteries must last for years. A gateway in the farm office collects the readings and forwards them to a cloud platform, where the owner views them on a mobile app.

### Questions

**1.** Would you choose LoRaWAN or NB-IoT for the field sensors? Justify your answer.

**2.** Which protocol would you use to send readings from the gateway to the cloud, and why?

**3.** What role does the gateway play in this design?

**4.** Why would IPv6 and 6LoWPAN be useful if the farm later grows to thousands of sensor nodes?

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### Expected Learning Outcome

*Students match communication technologies, protocols and networking concepts to the range, power and data needs of a realistic IoT deployment.*

## 📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

## 💭 Section N — Student Reflection

**What did I learn in this unit?**

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**What was difficult for me?**

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**What do I need to practice more?**

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**One important concept I learned:**

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## 🗝 Section O — Instructor Answer Key

*(For instructor use — covers all objective and subjective items in this worksheet.)*

### MCQ Answers

- 1. B — BLE is the low-energy variant of Bluetooth, ideal for battery-powered wearables and beacons.
- 2. C — Zigbee builds on IEEE 802.15.4, the low-rate wireless personal area network standard.
- 3. D — NFC works only when devices are almost touching, which is why it suits secure contactless payments.
- 4. A — LoRaWAN uses unlicensed sub-GHz bands, while NB-IoT and LTE-M run in licensed cellular spectrum.
- 5. B — LTE-M has higher throughput and mobility support than NB-IoT.
- 6. C — The broker is the central server that routes messages from publishers to subscribers by topic.
- 7. D — CoAP is a lightweight RESTful protocol that runs mainly over UDP for constrained devices.
- 8. A — WebSockets keep the connection open so client and server can send data at any time.
- 9. B — IPv6 addresses are 128 bits long, versus 32 bits for IPv4.
- 10. C — 6LoWPAN compresses headers and fragments packets so IPv6 can run on tiny low-power radios.

### Fill in the Blanks — Answers

- 1. publish-subscribe
- 2. gateway
- 3. NB-IoT
- 4. UDP
- 5. Wireless Sensor Network (WSN)

### True/False — Answers

- 1. True
- 2. False
- 3. False
- 4. True
- 5. False

### Match the Following — Answer Key

1 → b  2 → f  3 → g  4 → a  5 → e  6 → d  7 → c

### Short Answer — Key Points

- 1. Short-range links cover centimetres to roughly 100 m (e.g. BLE, Zigbee, NFC); long-range links cover kilometres with low power (e.g. LoRaWAN, NB-IoT).
- 2. The broker receives every published message, filters it by topic and forwards it to all subscribed clients, so publishers and subscribers never talk directly.
- 3. CoAP is lightweight, mainly uses UDP and has small headers for constrained devices; HTTP runs over TCP with larger headers and more overhead.
- 4. IPv6 has 128-bit addresses, giving billions of devices unique addresses that IPv4 (32-bit) cannot provide.
- 5. A gateway connects local IoT networks to the internet/cloud, translates between protocols, and can filter or aggregate data before sending it on.

## Descriptive Questions — Key Points

- 1. BLE: ~10–100 m, very low power, wearables. Zigbee: ~10–100 m, mesh, low data rate, home/industrial automation. Wi-Fi: ~50–100 m, high bandwidth, higher power. RFID: passive tags, short range, tracking. NFC: a few cm, secure tap payments. Present as a table.
- 2. LoRaWAN: km range, very low power, unlicensed, low data. NB-IoT: licensed cellular, deep indoor, low data. LTE-M: higher data rate, mobility, voice-capable. 4G/5G: high bandwidth and low latency for video, vehicles and critical control.
- 3. Publishers send messages to topics on a broker, which forwards them to subscribers; mention QoS 0/1/2. Compare: CoAP (UDP, REST), HTTP (request-response, heavy), AMQP (queued, reliable, enterprise), WebSockets (persistent full-duplex).
- 4. IPv4 (32-bit) vs IPv6 (128-bit); 6LoWPAN compresses IPv6 for 802.15.4; WSN nodes sense and relay data to a sink; gateways bridge networks; addressing identifies devices and routing finds the path for packets.

## Practical Activities — Expected Outputs

- 1. Sample logic: subscribers = {}; def subscribe(t, n): subscribers.setdefault(t, []).append(n); def publish(t, m): [print(n, "received", m, "on", t) for n in subscribers.get(t, [])]. Subscribing "app" and "display" to home/temperature and publishing 28 prints two lines.
- 2. (1) BLE — short range, very low power, phone-compatible. (2) LoRaWAN (or NB-IoT if cellular coverage exists) — km range, years of battery life, tiny payloads. (3) NFC — a few cm range, secure tap-to-open.
- 3. Topics: building/floor1/room101/temperature, building/floor1/room101/light, building/floor2/room201/temperature. Wildcard: building/+/+/temperature. QoS 0 for frequent routine readings; QoS 1 (or 2) for the high-temperature alert so it is not lost.

## Concept Map — Sample Answers

- Short-Range: BLE, Zigbee, Wi-Fi, NFC, RFID.  Long-Range: LoRaWAN, NB-IoT, LTE-M, 4G/5G.  Protocols: MQTT, CoAP, HTTP/HTTPS, AMQP, WebSockets.  Networking: IPv4/IPv6, 6LoWPAN, WSN, gateways, routing.

## Case Study — Key Points

- 1. Either is defensible. LoRaWAN suits remote farmland (long range, unlicensed spectrum, years of battery, private gateway); NB-IoT is better only if reliable cellular coverage exists and a subscription is acceptable.
- 2. MQTT — lightweight publish-subscribe suits small periodic readings and unreliable links; HTTPS is a workable alternative where simple REST calls are enough.
- 3. It collects data from the field network, translates LoRa messages into IP traffic, and forwards them securely to the cloud.
- 4. IPv6 gives each node a unique address as the network scales, and 6LoWPAN lets small low-power radios carry IPv6 efficiently.